



INDIAN OCEAN

The Indian Ocean is the world's third-largest ocean. Its northern ocean currents change direction according to the monsoon winds, flowing southwest along the coast of Somalia in the northern winter and the opposite direction in the northern summer.



Endangered green turtle

AREA: 26,064,036 sq miles (67,469,536 sq km)

Includes: Andaman Sea, Arabian Sea, Bay of Bengal, Great Australian Bight, Gulf of Aden, Gulf of Oman, Java Sea, Mozambique Channel, Persian Gulf, Red Sea, Timor Sea, Strait of Malacca

AVERAGE DEPTH: 12,720 ft (3,877 m)

DEEPEST POINT: 23,376 ft (7,125 m)
Java Trench

COASTLINE: 41,338 miles (66,526 km)

CLIMATE: Cool, dry winds blow over the ocean from the northeast between February and March. Between August and September, the wind changes direction and southwesterly winds blow north from the ocean, bringing heavy monsoon rain and flooding to coastal regions.

NATURAL RESOURCES: Oil and gas fields, sand and gravel, fish.

ENVIRONMENTAL ISSUES: Oil pollution in the Arabian Sea, Persian Gulf and Red Sea; endangered sea creatures include the dugong, turtles, and whales.



Oil production in the Arabian Sea



ARCTIC OCEAN

The Arctic Ocean is the world's smallest ocean. Between December and May, most of the ocean is covered by polar ice, which can be up to 100 ft (30 m) thick.

AREA: (3,351,811 sq miles (8,676,520 sq km)

Includes: Baffin Bay, Barents Sea, Beaufort Sea, Chukchi Sea, East Siberian Sea, Greenland Sea, Hudson Bay, Hudson Strait, Kara Sea, Laptev Sea, and the Northwest Passage

AVERAGE DEPTH: 6,349 ft (1,935 m)

DEEPEST POINT: 18,400 ft (5,680 m)

Fram Basin

COASTLINE: 28,204 miles (45,389 km)

CLIMATE: Polar, with continuous cold and narrow annual temperature ranges.

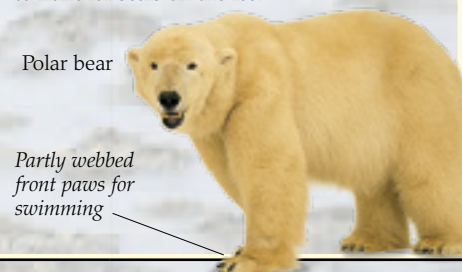


Ice breakers have thick steel hulls to crush ice and open up a lane for other ships

NATURAL RESOURCES: Oil and gas fields, sand and gravel aggregates, fish, and marine mammals.

ENVIRONMENTAL ISSUES: The extent of the sea ice is diminishing due to climate change creating problems for many marine animals. As the ice melts earlier, polar bears have less time to hunt for seals on the ice.

Polar bear



Partly webbed front paws for swimming



SOUTHERN OCEAN

The Southern Ocean is the world's fourth-largest ocean. Parts of the ocean freeze in the southern winter, forming the vast Ronne and Ross ice shelves. Currents beneath the ice shelves cause giant slabs of ice to break away, which melt as they float northwards.

AREA: 8,102,165 sq miles (20,973,318 sq km)

Includes: Amundsen Sea, Bellingshausen Sea, Ross Sea, Weddell Sea

AVERAGE DEPTH: 14,760 ft (4,500 m)

DEEPEST POINT: 23,737 ft (7,235 m)

South Sandwich Trench

COASTLINE: 11,165 miles (17,968 km)

CLIMATE: Polar, with continuous cold and narrow annual temperature ranges.

NATURAL RESOURCES: Probable large oil and gas fields, sand and gravel aggregates, fish, krill.

ENVIRONMENTAL ISSUES: Ultraviolet radiation penetrating through the ozone hole above the Antarctic is damaging phytoplankton. Although protected by the 1959 Antarctic Treaty and subsequent annexes, some illegal and unregulated fishing still occurs. However, protected whale and fur seal populations are making a comeback after overexploitation in the 18th and 19th centuries.

Characteristic flat-topped Antarctic iceberg

